

Evidence-First Bengali Hallucination Detection: A Deterministic Selective Cascade

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[Public reproducibility package](#)

Abstract—Our Bengali hallucination detector treats the benchmark as a mixture of tasks rather than one binary prompt. Source-lineage checks, structured solvers, context grounding, lexical and legal evidence, and relation-typed fact records decide high-confidence cases. A quantized open-weight model sees only the unresolved rows. Public macro-F1 rose from 0.657 to 0.967; the final Phase 1 result was private macro-F1 0.982 and rank 1. On the 299 released labels, the full cascade obtains 0.9933 macro-F1, while deterministic stages cover 225 rows without error. Three offline Kaggle runs on two Tesla T4 GPUs reproduced the scored 2,516-row CSV byte for byte in 440.87 to 593.43 seconds. A 5,000-row same-mix test took 14.60 minutes. Even an all-uncertain estimate stays below the nine-hour limit. The notebook computes every prediction from raw rows and pinned public assets; it does not load a saved prediction vector.

Index Terms—Bengali, hallucination detection, selective inference, evidence routing, reproducibility

I. TASK AND DESIGN

Each competition row contains an optional Bengali context, a prompt, and a model response. A fabricated, unsupported, or factually wrong response receives `label=0`; a faithful response receives `label=1`. The released development set contains 299 labeled rows, whereas Phase 1 contains 2,516 unlabeled rows and Phase 2 is expected to contain about 5,000 rows from broader sources [1], [2].

The rows do not form one uniform classification task. They include context entailment, arithmetic, multiple-choice questions, Bengali morphology, idioms, legislation, historical dates, literature, and civic information. A single prompt forces one model to solve all of these relations with the same evidence policy. Early experiments showed that this was accurate enough to establish a baseline but unstable across the released set and the public leaderboard.

We changed the decision rule to *selective resolution*. A rule or evidence source may emit a label only when its scope and relation are explicit. Everything else is deferred to a bounded LLM judge. The design has three practical goals:

- 1) preserve high recall for hallucinated responses, the `label=0` class;
- 2) avoid treating semantically related retrieval as ground truth;
- 3) reproduce the same CSV offline on Kaggle T4×2 within nine hours.

The resulting system couples a trust-aware deterministic router with a small, fixed inference tail. The router reduces

the cost and variance of LLM judging; the judge handles cases that do not fit a defensible exact rule.

II. SYSTEM

A. Normalization and Task Routing

The first stage normalizes Bengali and Latin digits, Unicode punctuation, spacing, answer aliases, and inline options. It also identifies whether context is absent, evidential, or merely instructional. This distinction matters: in several grammar examples the context describes the task rather than supplying the answer.

Exact lineage checks recover released BenHallu-style records and public question-answer sources only when normalized prompt, response, and relevant context fields agree. Reliability-weighted answer banks may resolve a row when their accepted answer relation is unambiguous. Approximate matches can propose a candidate, but they cannot overrule an exact source or an intrinsic context contradiction.

B. Structured Solvers

Small deterministic solvers handle arithmetic, numeric ranges, Bengali compound number words, multiple-choice serialization, morphology, compounds, and prefixes. Relation guards bind a date or number to the clause containing the requested entity and predicate. This prevents a valid year in the same passage from answering the wrong birth, death, publication, or establishment question.

For context rows, the system compares response claims with the supplied passage. Exact answer spans are accepted only inside a semicolon-aware clause that also contains prompt anchors. Named entities, dates, negation, and alternatives are checked before any external fact can contribute. The supplied context takes precedence over outside plausibility.

C. Evidence Hierarchy

Table I summarizes the admission policy. An authoritative exact lineage source can decide alone. Conflicting answer banks require weighted agreement. Free-form factual records require at least two independent public sources and an explicit relation type. Idiom records distinguish the figurative meaning requested by the prompt from a literal gloss. These restrictions were added after broader retrieval produced plausible but incorrectly related evidence.

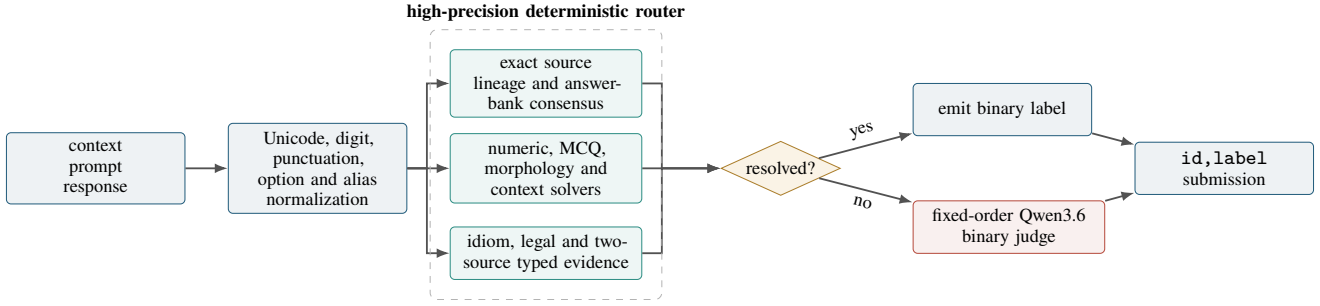


Fig. 1. Frozen v91 inference path. Deterministic stages label only relation-matched cases. The open-weight judge receives the unresolved tail, not the full test set.

TABLE I
EVIDENCE ADMISSION POLICY

Evidence family	Condition required to emit a label
Exact lineage	Prompt, response, and applicable context agree
Answer banks	Relation match and reliability-weighted consensus
Structured rule	Parsed operands, entity, and predicate all align
Context evidence	Same clause plus prompt anchors; no contradiction
Idiom or law	Exact headword/section and requested sense or provision
Free-form fact	Typed relation and two independent recorded sources
Fuzzy retrieval	Candidate generation only; never a stronger override

D. Conflict Resolution and Abstention

The router records a source family, relation, and confidence with each proposed decision. Intrinsic context contradiction has the highest priority because the task asks whether the answer follows the supplied passage. An exact authoritative source comes next, followed by relation-aligned structured parsing and consensus evidence. Fuzzy retrieval never decides a row by itself.

Conflicting proposals are not averaged blindly. If two sources claim different relations, both are discarded for that question. If they claim the same relation but disagree on the answer, the system requires a stronger exact source or abstains. This policy increased the judge queue in a few places, but it prevented weak retrieval from flipping labels that a trusted source had already resolved.

E. Selective Open-Weight Judge

After routing, 158 Phase 1 rows remain unresolved. They are sorted by row ID and processed in fixed batches. Context rows receive one context-NLI view. No-context rows receive up to three short views: factual verification, strict exam grading, and a zero-bias check. A faithful decision requires unanimous positive votes; evaluation stops after the first zero.

The judge uses Qwen3.6-27B Q4_K_P, a 16.332 GiB GGUF checkpoint, served by pinned llama.cpp b9637 across two T4 GPUs [6], [7]. Thinking is disabled. A grammar restricts output to one binary token. Seeds, row order, prompt order, and batches are fixed.

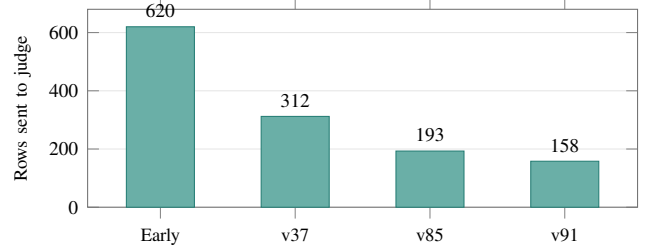


Fig. 2. The deterministic router reduced the Phase 1 judge queue by 74.5%.

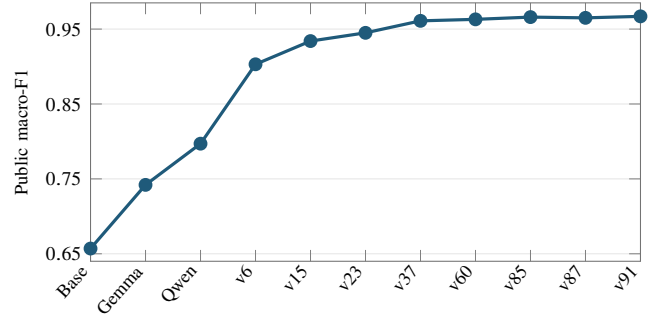


Fig. 3. Public leaderboard progression. The largest gain followed the shift from uniform judging to source and task routing.

III. EXPERIMENTS

A. Evaluation Protocol

We used the 299 released labels for validation and error analysis, not as proof that a large fine-tune would transfer. We tracked macro-F1, F1 for each class, accuracy, and deterministic coverage. Context and no-context rows were inspected separately. Candidate rules were admitted only when they expressed a reusable relation and passed the regression suite.

The public leaderboard exposed roughly 30% of Phase 1. A rounded score change therefore could not reveal a specific row label. We treated leaderboard results as aggregate evidence and retained the last reproducible anchor whenever a proposed correction regressed.

B. Scored Iterations

Table II reports representative scored versions. The initial character and embedding models established a 0.657 to 0.663

TABLE II
REPRESENTATIVE LEADERBOARD ITERATIONS

Version	Public F1	Principal change
Baseline	0.657	character and embedding models
Gemma	0.742	12B QAT generic judge
Qwen	0.797	27B lexical/factual judge
v6	0.903	source and structured routing
v15	0.934	expanded lineage, calibrated rules
v23	0.945	provenance repair batch
v37	0.961	source-lineage audit
v60	0.963	conservative factual corrections
v85	0.966	reproducible two-source anchor
v87	0.965	plausible fact edits; reverted
v91	0.967	typed idiom and relation guards

TABLE III
RELEASED-LABEL EVALUATION

Metric	Full cascade	Deterministic subset
Rows	299	225
Macro-F1	0.993264	1.000000
F1, hallucinated (label=0)	0.992701	1.000000
F1, faithful (label=1)	0.993827	1.000000
Accuracy	0.993311	1.000000

floor. Gemma 12B QAT and the Qwen judge improved factual reasoning, but remained prompt-sensitive. v6 introduced source and structured routing and jumped to 0.903. Later gains came from lineage audits, stricter provenance, legal evidence, and relation guards rather than a larger model.

v91 changed five evidence-backed decisions from v85. Its public gain was 0.001, but the final private leaderboard score was 0.982, rank 1. This gap is consistent with the limited public split and is not presented as evidence that every v91 change generalized.

C. Failed Approaches

A prompt-vote ensemble obtained 0.8512 locally and 0.783 publicly. It overfit prompt behavior on the small released set. A compact TF-IDF verifier improved from 0.7879 to 0.8261 out of fold, but remained weak on no-context cultural facts. Raw external-QA retrieval often returned adjacent passages with the wrong relation or authority. A global context substring rule failed on instructional grammar contexts. Finally, v87 applied two externally plausible fact edits and fell from 0.966 to 0.965. We reverted it and tightened the admission policy.

After those runs, retrieval became a proposal mechanism, context rules became relation-aware, and each researched evidence record required a source and typed predicate. Candidate batches without a clear expected direction were quarantined instead of submitted.

D. Released-Label Results

The two remaining errors are internally inconsistent released controls. We left them unchanged rather than add row-specific exceptions.

TABLE IV
FROZEN DEPLOYMENT MEASUREMENTS

Property	Frozen result
Hardware	Kaggle Tesla T4×2; internet disabled
Model	Qwen3.6-27B Q4_K_P, 16.332 GiB
Phase 1 runs	440.87, 527.03, and 593.43 seconds
5,000 same-mix run	876.14 seconds, or 14.60 minutes
Pessimistic bound	all rows, three views: about 5.2 to 5.5 hours
Output	2,516 IDs; 1,181 zeros and 1,335 ones
SHA-256	0fec5d610f4e7b94...
Tests	249 repository tests passing at freeze

IV. REPRODUCIBILITY AND DEPLOYMENT

The inference notebook reads the literal organizer path:

```
/kaggle/input/competitions/  
bengali-hallucination/test_set.csv
```

It does not search for alternate filenames, branch on the row count, call an external API, or load a Phase 1 prediction vector. The organizers can replace the file mounted at that path for Phase 2 without editing a notebook variable.

Three notebook runs produced byte-identical files matching competition submission 54836339. The full digest is published in the freeze manifest. The public package pins the notebook version, Qwen checkpoint, CUDA runtime, evidence dataset, hardware, and expected hash.

A. Determinism Controls

Reproduction depends on more than a fixed random seed. The notebook sorts unresolved IDs before scheduling work and uses fixed batch boundaries. It starts both llama.cpp servers with pinned command lines, waits for health checks, and assigns the same prompt view to the same server on every run. Generation is limited to one grammar-constrained token. The CSV writer preserves incoming IDs, writes integer labels, and uses a fixed column order. A final validation cell checks row count, sequential Phase 1 IDs when applicable, nulls, label values, counts, and SHA-256.

The SHA check is a Phase 1 reproduction check, not a Phase 2 condition. On the held-out file, the notebook validates schema and IDs but does not expect the Phase 1 digest or label distribution. This separation prevents the notebook from recognizing the Phase 1 file and switching to a stored result.

B. Phase 2 Procedure

The frozen notebook infers the row count and preserves incoming IDs. Before submission, we duplicated the current distribution to roughly 5,000 rows and verified schema, row-count independence, offline execution, and GPU memory. A same-mix run took 14.60 minutes. Because broader Phase 2 data may route fewer cases deterministically, we also estimated the case in which all 5,000 rows need all three LLM views. The resulting 5.2 to 5.5 hour bound remains below the nine-hour budget.

TABLE V
PHASE 2 ROUTING UNDER THE ANNOUNCED SOURCE MIX

Source family	Primary handling and abstention condition
Common Crawl and news	Context grounding; defer unsupported outside claims
Wikipedia and Banglapedia	Exact entity-relation match; no approximate override
Government services	Context clauses, dates, fees, offices, and named entities
Bangladesh laws	Exact act and section when present; otherwise judge
NCTB textbooks	MCQ, numeric, grammar, and context solvers
Literature sites	Exact work/author relation or reliability consensus
Unknown source	Skip lineage and use the fixed judge path

C. Expected Domain Shift

The organizer preview names Common Crawl, Bengali Wikipedia, Banglapedia, Bangladesh government pages, recent newspapers, legislation, NCTB textbooks, and Bengali literature sites [2]. Phase 1 source lineage should therefore cover a smaller share of Phase 2. The system does not turn an unfamiliar source into an approximate lineage match. When exact source and relation checks fail, the row moves to context reasoning or the judge.

This fallback is deliberate. A loss of deterministic coverage increases runtime but is safer than forcing Phase 1 evidence onto new domains. The all-uncertain bound measures that failure mode directly. It also explains why the notebook keeps the 27B judge even though only 158 Phase 1 rows required it.

D. Method and Rule Disclosure

The router contains short citation-backed factual records researched from public sources for unlabeled prompts. It never used hidden labels, manually assigned row labels, paid APIs, or closed hosted inference. We told the organizers when exact prompt/source relationships were discovered and how the public-source records were built. They replied that Kaggle foundational rules were not organizer-imposed and that a solution without manually labeled rows, API calls, or other rulebook violations would not be disqualified. The pipeline still computes every output label from the provided row and its pinned evidence.

The public repository omits competition test data, predictions, credentials, and model weights. It includes the notebook, report, version manifest, citations, data-source register, third-party notices, form values, and run instructions. Model and runtime files remain on exact public Kaggle versions.

E. Limitations

The private 0.982 result does not guarantee equal performance on broader Phase 2 sources. Exact lineage helps reproduce known source families but cannot replace general reasoning on new civic, legal, textbook, and news material. The 299 labels are too small to establish performance for every domain. Public-source records also require continuing license and provenance review. In a deployed civic system, unresolved

TABLE VI
PUBLISHED ARTIFACT BOUNDARY

Location	Contents
GitHub	Notebook, LaTeX report, PDF, manifests, citations, and runbook
Kaggle Model	Pinned public Qwen3.6-27B quantization
Kaggle Dataset	Router evidence, llama.cpp CUDA build, notices
Competition mount	Organizer-provided CSV at the literal required path
Notebook output	Newly computed <code>submission.csv</code> only
Not published	Competition rows, scored predictions, credentials

legal or service information should be escalated rather than accepted from a binary detector alone.

V. CONCLUSION

Changing the unit of reasoning mattered more than enlarging the judge. Exact provenance and typed solvers handled cases with clear relations; a fixed open-weight model handled the remaining ambiguity. The result was more accurate than uniform LLM judging, fast enough for offline T4×2 inference, and reproducible byte for byte from raw input. Each deterministic label identifies its evidence family, and every unresolved label follows a fixed prompt and voting path.

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